Flipkart Customer Service Analysis

Milestone 3 Report: Evaluating Support Operations & Impact on Customer Retention

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Executive Summary

Customer service is a major factor in keeping customers loyal and protecting brand value at Flipkart. When shoppers face problems with orders, payments, or billing, their interaction with customer support often decides whether they will shop with Flipkart again. This report presents an in-depth analysis of **30,000 customer service interactions** across four contact channels: Call-Center, Chatbot, Email, and Web support.

By analyzing this dataset, we uncovered four important business findings that directly impact customer retention:

- **63% Feedback Gap (Low Response Rate):** Only **37% of customers submitted a CSAT rating** (Response Rate), while 63% left no feedback. This severe gap in feedback collection makes it difficult to track overall customer happiness accurately across the entire customer base.
- **51.75% Overall Dissatisfaction:** Among customers who submitted feedback, overall sentiment is strongly negative. More than half of all interactions were rated as either *Negative* (33.67%) or *Very Negative* (18.08%).
- **Similar Frustration Across All Channels:** Dissatisfaction is not limited to one channel. Customers report almost identical negative sentiment across Call-Center, Chatbot, Email, and Web channels (around 51% to 53% negative sentiment on each channel).
- **Resolution Quality Matters More Than SLA Speed:** Testing our business hypotheses showed that simply achieving higher **SLA Compliance** (handling calls within the promised SLA timeframe) or keeping calls short does not improve CSAT scores. Customers are unhappy because of underlying product, billing, and payment issues—not just the duration or answer speed of the support call.

30,000	5.54 / 10	25.0 min	37%	62%
TOTAL CALLS	AVERAGE CSAT	AVERAGE DURATION	RESPONSE RATE	SLA COMPLIANCE

Project Overview

Flipkart is one of India's leading e-commerce marketplaces, offering millions of products across electronics, fashion, home goods, and daily essentials. In online retail, customers expect smooth ordering, accurate billing, and reliable delivery. When issues occur, customer support must resolve them quickly and fairly.

Recently, Flipkart noticed a drop in customer retention rates. To find out if customer service challenges are contributing to this decline, this project analyzes operational support data. By examining response times, contact

channels, regional call centers, and reasons for customer contact, this study identifies the exact friction points that hurt customer satisfaction and cause churn.

Business Problem & Objective

Business Problem: Flipkart has noticed a decline in customer retention. The company needs to identify whether customer service operations are causing customers to leave, and which specific areas—such as response times, call centers, contact channels, or reason for calling—need urgent improvement.

Business Objective: The primary goals of this Milestone 3 report are to:

1. Define key customer service performance metrics and illustrate how they connect to customer retention.
2. Test five specific business hypotheses using descriptive statistics and pivot tables from 30,000 call records.
3. Evaluate the performance of four contact channels and four regional call centers.
4. Provide clear, data-backed recommendations in simple English to improve customer satisfaction and reduce churn.

Project Scope

The scope of this project covers the analysis of **30,000 customer support records** collected during October 2020. The dataset includes four customer contact channels (Call-Center, Chatbot, Email, and Web) and four operational call center locations (Delhi, Mumbai, Kolkata, and Chennai).

> **Important Note on Customer Retention Proxy**
>
> The raw data file contains interaction-level records but does not have a direct repeat-purchase or customer churn tracking column. Therefore, **Customer Satisfaction (CSAT) score and Customer Sentiment are used as direct proxies for retention**. In e-commerce, customers who express Very Negative or Negative sentiment and award low CSAT scores have the highest risk of stopping shopping with the platform.

Dataset Overview

The dataset contains 30,000 rows and 13 original columns, capturing details about customer identity, contact reason, response speed, handling duration, and customer feedback.

Column Name	Data Type	Description
ID	Text	Unique tracking ID for each customer service interaction.
customer_name	Text	Name of the customer who contacted support.
Gender	Text	Gender of the customer (m / f).
sentiment	Text	Customer sentiment rating (Very Negative, Negative, Neutral, Positive, Very Positive).

Column Name	Data Type	Description
csat_score	Number / Text	Customer satisfaction score from 1 to 10, or blank if not rated.
call_timestamp	Date / Time	Exact date and time when the interaction took place.
reason	Text	Primary reason for contacting support (Billing Question, Payments, Service Outage).
city / state	Text	Location of the customer.
channel	Text	Support channel used (Call-Center, Chatbot, Email, Web).
response_time	Text	Service level agreement (SLA) status (Within SLA, Above SLA, Below SLA).
call duration in minutes	Number	Total time taken to handle the interaction in minutes.
call_center	Text	Support center location handling the call (Delhi, Mumbai, Kolkata, Chennai).

Data Cleaning & Feature Engineering

Before analyzing the data, a thorough data cleaning and preparation process was performed in Excel. The key steps are documented in the table below:

Data Quality / Step	Observation in Raw Data	Action Taken & Purpose
Duplicate Record IDs	No duplicates found.	All 30,000 interaction IDs were confirmed unique.
Missing CSAT Scores	~63% of rows had blank CSAT cells.	Blank cells were labeled as **"No Rating"**. We did not replace them with averages so that true customer satisfaction scores remained accurate.
Missing Customer Info	Some blank names, gender, and city/state.	Retained in the dataset so we do not lose call volume counts. Missing geography was labeled as #N/A.
Added: Call_day	Derived from `call_timestamp`.	Extracted the day of the week (e.g., Monday, Sunday) to analyze if call volumes or CSAT scores change on specific days.
Added: Has_CSAT	Binary flag (Yes / No).	Created a simple Yes/No column to track whether a customer gave a CSAT rating. Used to measure Response Rate (37%).
Added: CSAT Bucket	Grouped CSAT scores.	Categorized scores into **Low (1–3)**, **Medium (4–6)**, **High (7–10)**, and **NA** for unrated calls.
Added: Duration Bucket	Grouped call duration in minutes.	Categorized duration into **Short (≤15 min)**, **Medium (16–30 min)**, and **Long (>30 min)** to test efficiency.

Metric Tree

To clearly illustrate how frontline operational actions impact the overarching business objective, we designed a structured **Metric Tree**. This visual diagram maps the primary goal of customer retention into two core performance branches: **Customer Satisfaction** and **Operational Efficiency**.

OVERALL BUSINESS GOAL: Improve Flipkart Customer Retention

Customer Satisfaction Metrics (Loyalty Proxy)

- **Average CSAT Score:** 5.54 / 10
- **Response Rate (CSAT Capture Rate):** 37% (63% missing)
- **Negative Sentiment Share:** 51.75% combined
- **Positive Sentiment Share:** 21.57% combined

Operational Efficiency Metrics (Speed & Workload)

- **SLA Compliance Rate:** 62% Within SLA
- **Average Call Duration:** 25.0 minutes
- **Total Support Volume:** 30,000 interactions
- **Contact Reason Load:** 72% Billing Questions

KPI Definitions

The table below defines each key metric used in our dashboard and hypothesis testing, clearly separating feedback capture metrics from SLA responsiveness metrics:

Metric Name	Definition & Formula	Target Benchmark	Why It Matters for Flipkart
Total Calls	Total count of support interactions logged.	30,000 calls	Shows total support workload and customer demand.
Average CSAT	Mean satisfaction score among customers who gave a rating (1 to 10 scale).	≥ 8.0 / 10	Direct measure of customer happiness after speaking with support.
Response Rate (CSAT Capture Rate)	**Percentage of customers who submitted a CSAT rating** (Rated Calls / Total Calls).	≥ 70%	Quantifies feedback coverage; currently at **37%**, indicating a 63% blind spot.
SLA Compliance (SLA Adherence)	**Percentage of calls handled within SLA** (Within SLA Calls / Total Calls).	≥ 85% Within SLA	Measures speed of service and adherence to response time SLAs (currently **62%**).
Average Duration	Mean call duration in minutes across all calls.	15 – 20 minutes	Shows how much time is needed to solve a customer problem (currently **25.0 min**).

Metric Name	Definition & Formula	Target Benchmark	Why It Matters for Flipkart
Negative Sentiment %	Share of calls tagged as Negative or Very Negative sentiment.	≤ 20%	High negative sentiment warns of customer frustration and churn risk (currently **51.75%**).

Hypotheses Analysis

We tested five structured business hypotheses using Excel pivot tables to understand what drives customer satisfaction. Each hypothesis was evaluated strictly on data evidence and assigned a clear status: **Not Proven** or **Partially Proven**.

H1: Above SLA calls are concentrated in specific Call Centers or Channels, and cause lower CSAT and higher Negative Sentiment.

NOT PROVEN

Metrics Tested: SLA Compliance (Within SLA / Above SLA), Average CSAT, Negative Sentiment %

Data Analysis & Result: Pivot table results show that Above SLA calls are evenly distributed across call centers and channels. Furthermore, the Average CSAT score is almost identical across all SLA categories: **Above SLA is 5.60**, **Below SLA is 5.57**, and **Within SLA is 5.52**. Because satisfaction scores do not drop when SLAs are breached, the hypothesis is not supported.

Business Recommendation: Do not focus only on SLA timer compliance. Investigate other factors such as issue resolution quality, agent helpfulness, and billing policy fairness to improve customer satisfaction.

H2: Reducing average response time (shifting calls from Above SLA to Within SLA) will improve CSAT scores.

NOT PROVEN

Metrics Tested: SLA Compliance, Average CSAT, Channel Sentiment

Data Analysis & Result: Average CSAT is almost the same across all SLA compliance categories (5.52 to 5.60) and across contact channels (Call-Center 5.61, Web 5.61, Email 5.49, Chatbot 5.46). Customer sentiment is also nearly identical across channels, with Negative + Very Negative sentiment at ~51% to 53% on every channel. Shifting response times alone does not change customer satisfaction.

Business Recommendation: Improve satisfaction by training agents on issue resolution quality and clear communication rather than pushing for speed alone.

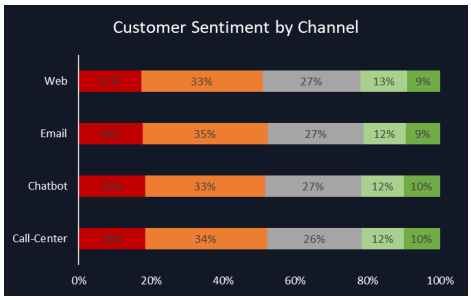


Figure 1: Customer Sentiment by Channel showing consistent ~51–53% negative sentiment across all four channels.

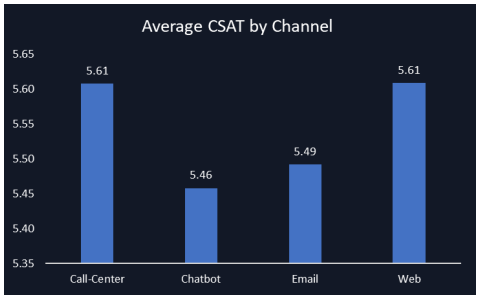


Figure 2: Average CSAT by Channel showing very similar satisfaction ratings across channels (5.46 to 5.61).

H3: Longer call duration is associated with lower CSAT and more negative sentiment, suggesting inefficient processes hurt satisfaction. NOT PROVEN

Metrics Tested: Duration Bucket (Short, Medium, Long), Average CSAT, Sentiment Distribution

Data Analysis & Result: Average CSAT is very similar across all call lengths: **Short calls average 5.49**, **Medium calls average 5.60**, and **Long calls average 5.51**. Sentiment distribution is also steady across Short, Medium, and Long interactions. Customers do not penalize Flipkart just because a call takes longer, provided the problem is being handled.

Business Recommendation: Focus on improving first-contact resolution quality rather than forcing agents to end calls quickly.

H4: Chatbot interactions have lower CSAT and more negative sentiment than human Call-Center channels, indicating a chatbot service gap. PARTIALLY PROVEN

Metrics Tested: Channel, Average CSAT, Sentiment Distribution

Data Analysis & Result: Chatbot interactions have the lowest Average CSAT score (**5.46** compared to **5.61** for Call-Center). However, its negative sentiment share is similar to other channels (18.51% Very Negative and 33.05% Negative). Because CSAT is slightly lower but overall sentiment is similar, the hypothesis is partially supported.

Business Recommendation: Improve the chatbot experience by upgrading automated issue resolution and providing a faster, smoother escalation path to human agents for complex queries.

H5: Billing Question calls (highest volume) drive a disproportionate share of negative sentiment and low CSAT compared to other reasons. PARTIALLY PROVEN

Metrics Tested: Reason, Total Calls, Average CSAT, Sentiment Distribution

Data Analysis & Result: As shown in Figure 3, **Billing Question is by far the highest-volume call reason at 72%** (21,410 calls), while Payments and Service Outage account for 14% each. However, Billing Question does not have the lowest CSAT (it averages **5.53** vs. **5.51** for Service Outage and **5.63** for Payments). Payments actually has the highest Very Negative sentiment (18.86%). The hypothesis is partially supported because Billing Questions affect the largest number of customers.

Business Recommendation: Prioritize simplifying invoice and billing explanations because they impact 72% of all callers, while also fixing payment failure workflows to reduce high frustration.

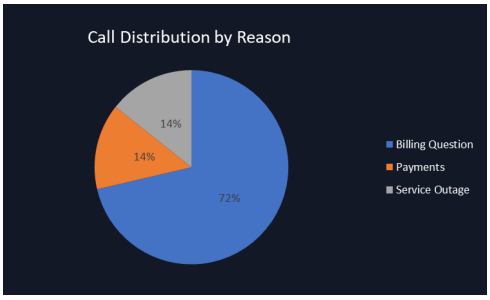


Figure 3: Call Distribution by Reason highlighting that Billing Questions account for 72% of all support inquiries.

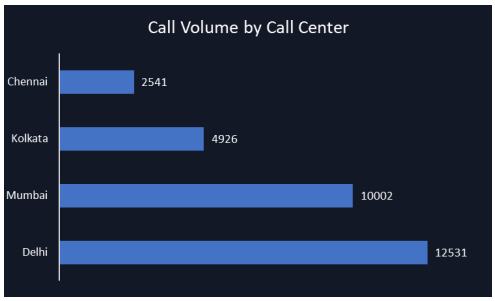


Figure 4: Call Volume by Call Center showing Delhi (12,531 calls) and Mumbai (10,002 calls) handling 75% of total volume.

Dashboard Walkthrough

The complete interactive Excel dashboard built for this project is presented below. It brings together all KPIs, sentiment distributions, regional volumes, and channel comparisons into a single management view:

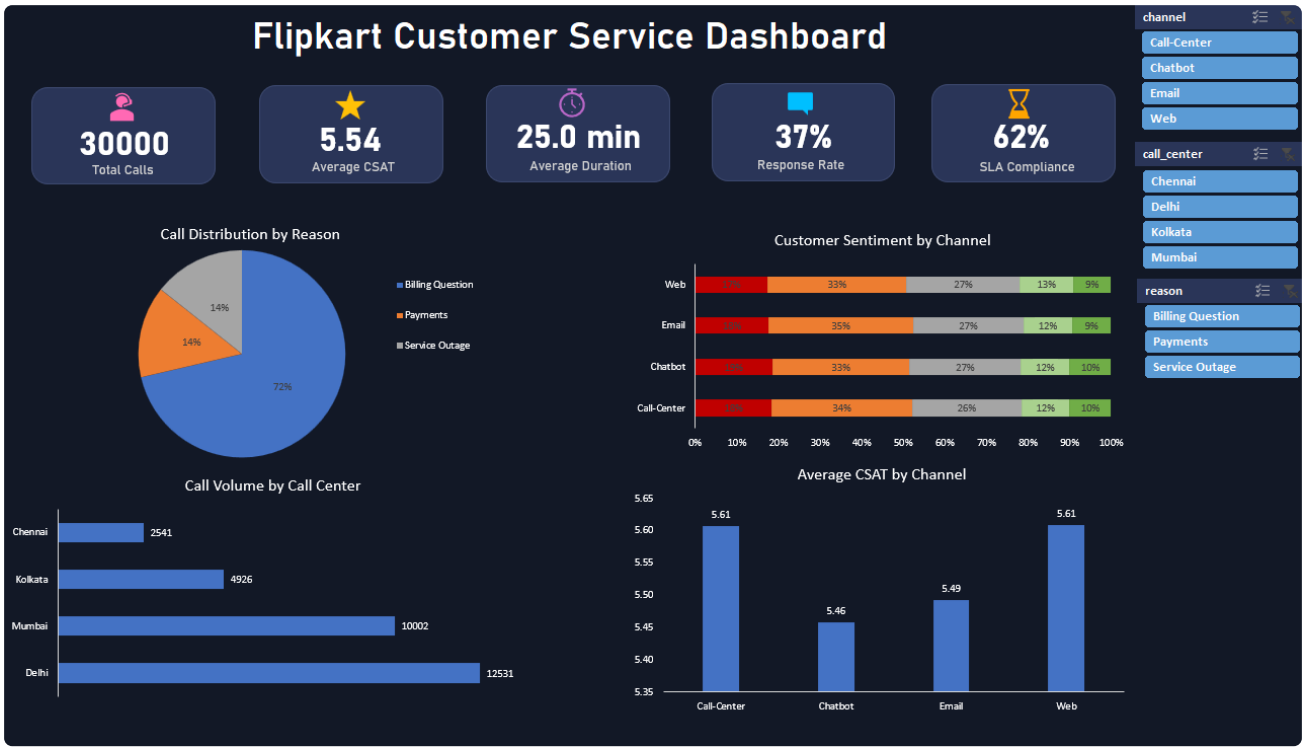


Figure 5: Full Flipkart Customer Service Dashboard showcasing overall volume, satisfaction, SLAs, reasons, and regional performance.

Detailed Dashboard Explanation:

- **Top KPI Summary Cards:** The top row immediately shows leadership the big picture: **30,000 Total Calls** handled, an overall **Average CSAT of 5.54** out of 10, an **Average Duration of 25.0 minutes**, a low **Response Rate of 37%** (percentage of customers who submitted a CSAT rating, meaning 63% left no feedback), and an overall **SLA Compliance of 62%** (percentage of calls handled within SLA).
- **Call Distribution by Reason (Pie Chart):** This chart shows that **72% of customers contact support for Billing Questions**, while Payments and Service Outages each generate 14% of volume. Fixing billing clarity is Flipkart's biggest opportunity to reduce support ticket volume.
- **Call Volume by Call Center (Horizontal Bar Chart):** This view compares the four regional support centers. **Delhi is the largest center with 12,531 calls**, followed closely by **Mumbai with 10,002 calls**. Together, Delhi and Mumbai handle over 75% of Flipkart's customer service workload. Kolkata (4,926 calls) and Chennai (2,541 calls) handle the remaining volume.
- **Customer Sentiment by Channel (Stacked Bar Chart):** This breakdown compares sentiment across Web, Email, Chatbot, and Call-Center. All channels show a red/orange band representing ~51% to 53% negative sentiment. This proves that customers experience similar frustration whether they speak to a human or use digital support.
- **Average CSAT by Channel (Column Chart):** This chart compares satisfaction scores across channels. Call-Center and Web lead slightly at **5.61**, Email is **5.49**, and Chatbot is lowest at **5.46**.

Key Business Insights

Based on our data analysis and hypothesis testing, we identified five major insights for Flipkart leadership:

- **The 63% Feedback Blind Spot (Low Response Rate):** Because the **Response Rate is only 37%** (percentage of customers submitting a CSAT rating), Flipkart is making customer retention decisions based on a minority sample of its users. Dissatisfied customers who leave without rating may be churning silently.
- **High Baseline Dissatisfaction:** Across the entire dataset, 51.75% of customer sentiment ratings are Negative or Very Negative. This indicates that more than half of contacting shoppers experience frustration during support.
- **Billing is the Main Driver of Support Workload:** Billing Questions represent 72% of all customer calls. Even small improvements in invoice clarity or automated billing explanations could deflect thousands of calls.
- **Speed Does Not Equal Happiness (SLA Compliance):** Looking at **SLA Compliance** (percentage of calls handled within SLA), customers whose calls were answered "Above SLA" had an average CSAT of 5.60, while calls answered "Within SLA" had an average CSAT of 5.52. This proves that customers care far more about getting their problem fixed than about how fast the agent answers.
- **Chatbot Needs Better Problem Resolution:** While chatbot volume is high, its CSAT score (5.46) is the lowest among all channels, showing that automated self-service currently fails to resolve customer issues satisfactorily.

Business Recommendations

To improve customer satisfaction and protect customer retention, Flipkart should implement the following five actionable strategies:

Recommendation Area	Actionable Strategy for Flipkart	Expected Business Benefit
1. Improve Feedback Collection (Raise Response Rate)	Introduce simple 1-click survey buttons (e.g., "Was this helpful? Yes/No") immediately after a chat or call ends. Send short SMS/WhatsApp reminders for unrated calls.	Raises **CSAT Response Rate** from 37% to >70%, giving managers a true picture of customer happiness.
2. Simplify Billing & Invoicing (72% of Volume)	Redesign the order invoice and billing explanation screen on the Flipkart app so pricing, discounts, and refunds are easy to understand without calling support.	Deflects up to 30% of Billing Question calls, reducing overall support costs.
3. Focus on Resolution Quality, Not SLA Speed	Shift agent training from "handle calls fast" to "first-contact resolution." Reward agents who solve customer issues completely on the first call rather than only targeting SLA timer compliance.	Improves CSAT scores above 5.54 and builds customer trust and retention.
4. Upgrade Chatbot & Human Escalation	Improve chatbot answers for simple billing and tracking questions, and add an easy "Talk to an Agent" button when a customer asks a complex question.	Lifts Chatbot CSAT from 5.46 toward 6.0+ and reduces customer frustration.

Recommendation Area	Actionable Strategy for Flipkart	Expected Business Benefit
5. Balance Regional Call Center Workloads	Route extra calls from high-volume centers (Delhi and Mumbai, which handle 75% of calls) to Kolkata and Chennai during peak shopping hours.	Improves **SLA Compliance** (percentage of calls handled within SLA) and ensures smoother workload balance across teams.

Limitations of Dataset

While this analysis provides valuable operational insights, the dataset has three specific limitations:

- **No Repeat-Purchase or Customer ID Column:** The dataset logs individual support tickets but does not include unique customer identifiers. Therefore, we cannot directly measure long-term customer churn or repeat purchase rates after a support call.
- **Missing CSAT Data (63% Unrated):** Because 63% of calls lack a CSAT score (a Response Rate of only 37%), the reported average score of 5.54 is based only on the minority of customers who chose to leave feedback.
- **No Cost or Staffing Data:** The data does not contain cost-per-call or agent headcount numbers, which limits our ability to calculate the financial cost of SLA breaches.

Future Improvements

For future analytical milestones, Flipkart data science and business teams should consider:

- **Linking Customer Support Data to Purchase History:** Merge support ticket logs with customer order databases to measure exactly how many customers stop shopping after a "Very Negative" support call.
- **Tracking First-Contact Resolution (FCR):** Add an FCR tracking column to see if repeat calls for the same issue cause lower CSAT scores.
- **Agent-Level Quality Scoring:** Record agent training levels and quality scores to identify top-performing support teams and share best practices across Mumbai, Delhi, Kolkata, and Chennai.

Conclusion

This Milestone 3 project evaluated Flipkart's customer service operations to understand why customer retention has been declining. Our analysis of 30,000 interactions revealed that while Flipkart maintains 62% SLA Compliance (calls handled within SLA) and a 25-minute average call duration, **51.75% of customer sentiment remains negative**, and **63% of calls lack feedback ratings** (37% Response Rate).

Crucially, hypothesis testing demonstrated that response speed and call duration alone do not determine customer happiness. Instead, customers are frustrated by underlying transactional issues—especially Billing Questions (72% of volume) and Payment failures. By simplifying billing clarity, improving chatbot escalation paths, and focusing on first-contact resolution quality, Flipkart can turn customer service into a driver of long-term loyalty and retention.

Appendix: Complete Data Dictionary

The table below lists all original data columns from the Flipkart customer service dataset along with the engineered features created during data preparation:

Column Name	Data Type	Source	Description & Categories
`ID`	String	Raw Data	Unique tracking number for each customer service interaction.
`customer_name`	String	Raw Data	Name of the customer contacting Flipkart support.
`Gender`	Categorical	Raw Data	Customer gender (`m` / `f`).
`sentiment`	Categorical	Raw Data	Customer sentiment rating (Very Negative, Negative, Neutral, Positive, Very Positive).
`csat_score`	Numeric / Text	Raw Data	Customer satisfaction score from 1 to 10, or "No Rating" if missing.
`call_timestamp`	Date/Time	Raw Data	Date and time when the interaction took place.
`reason`	Categorical	Raw Data	Primary contact reason (Billing Question, Payments, Service Outage).
`city` / `state`	String	Raw Data	Customer geographical location.
`channel`	Categorical	Raw Data	Support channel used (Call-Center, Chatbot, Email, Web).
`response_time`	Categorical	Raw Data	SLA performance rating (Within SLA, Above SLA, Below SLA).
`call duration in minutes`	Numeric	Raw Data	Total length of the interaction in minutes.
`call_center`	Categorical	Raw Data	Regional call center location (Delhi, Mumbai, Kolkata, Chennai).
`Call_day`	String	Engineered	Day of the week extracted from timestamp.
`has_CSAT`	Categorical	Engineered	Yes/No flag showing whether a CSAT rating was provided. Used for Response Rate (37%).
`csat_bucket`	Categorical	Engineered	Score grouping: Low (1–3), Medium (4–6), High (7–10), or NA.
`duration_bucket`	Categorical	Engineered	Duration grouping: Short (1–15 min), Medium (16–30 min), Long (31–45 min).